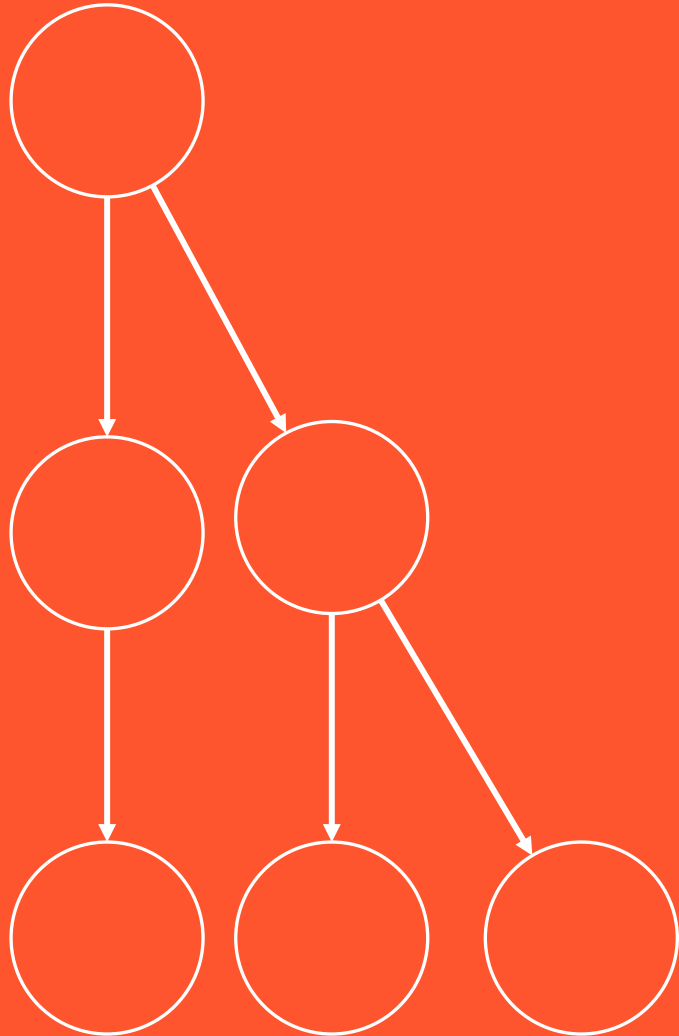


BCOG 100

Comparative Approaches

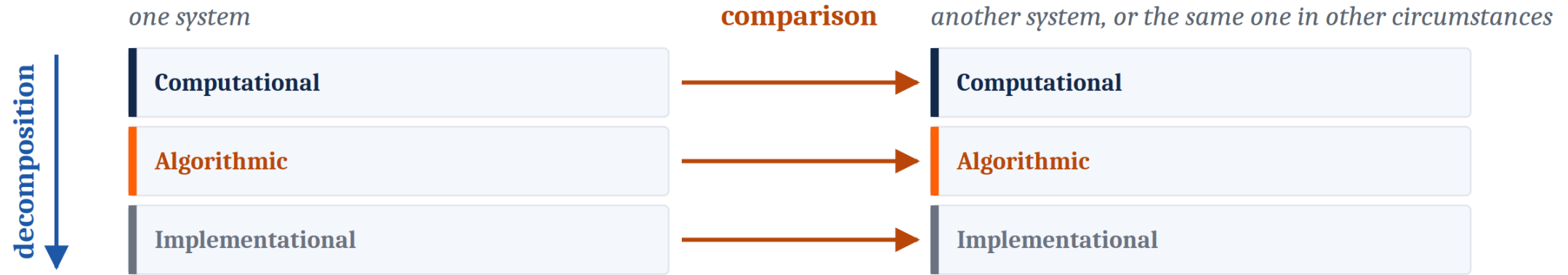
Lecture 2.1 — The Comparative Method



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A Second Method, at Right Angles to the First

Two methods, at right angles.



Marr goes down. Comparison goes across. A field with both can ask what a capacity is, and what else in the world has anything like it.

A field with both can ask what a capacity **is**, and also ask **what else in the world has anything like it**.

The practice has a name: the **comparative method**.

One Figure, Two Methods

Down a column is **Marr**. Across a row is **comparison**.

The frog is **not** a poor version of a human visual system. It is solving a different problem, and its machinery fits that problem well.

	Human vision	Frog vision
Computational <i>What problem is being solved?</i>	Recover a general description of surfaces and objects — one that can serve many tasks at once	Find a small dark thing moving as an insect moves, and strike; detect a spreading shadow, and flee
Algorithmic <i>By what procedure?</i>	Build the description in stages, from local edges toward viewpoint-stable objects	Compute a few specific features and route them almost directly to action
Implementational <i>In what physical stuff?</i>	Retina, a thalamic relay, and a large cortical hierarchy	Much of the work is done in the retina itself; output goes chiefly to the optic tectum in the midbrain

What Would You Conclude?



A person and an image-recognition model both label the wolf as a dog.

Across a large test set:

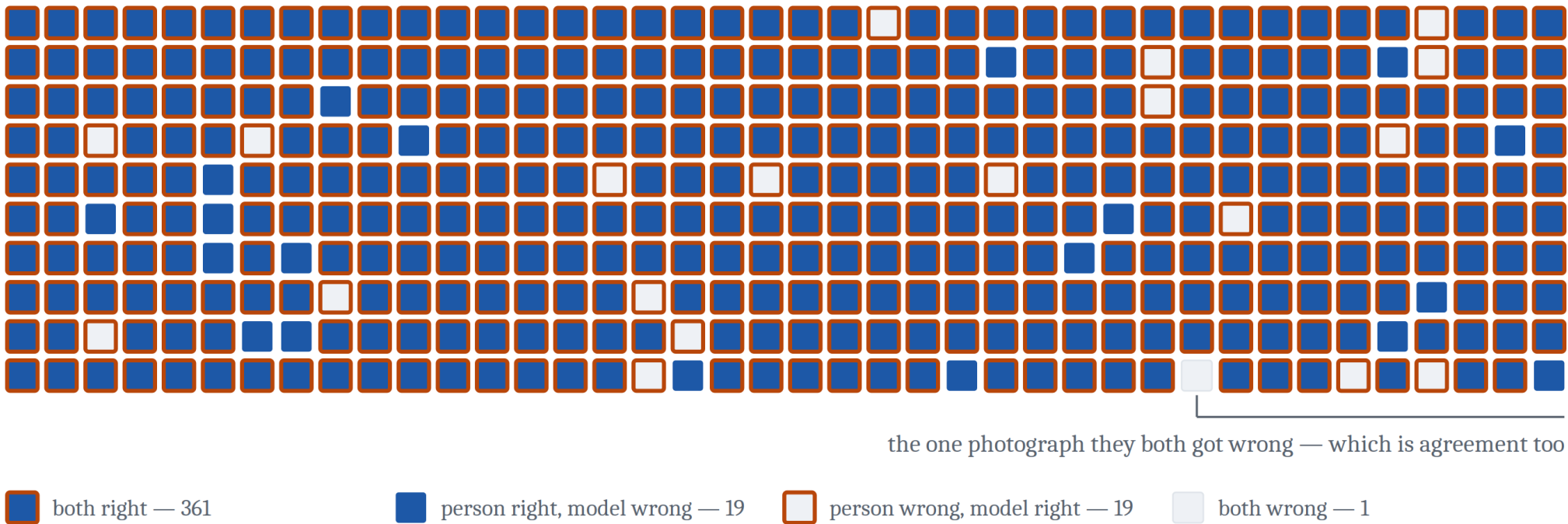
- the person is right about **95%** of the time
- the model is right about **95%** of the time
- the two **agree with each other 95%** of the time

This is the kind of finding reported as evidence that the model sees the way people do.

Does it show that?

A Comparison Cannot Interpret Itself

Fill = the person got it right. Thick border = the model got it right.
Each marks 95% of the same 400 photographs, and the two have nothing to do with each other.

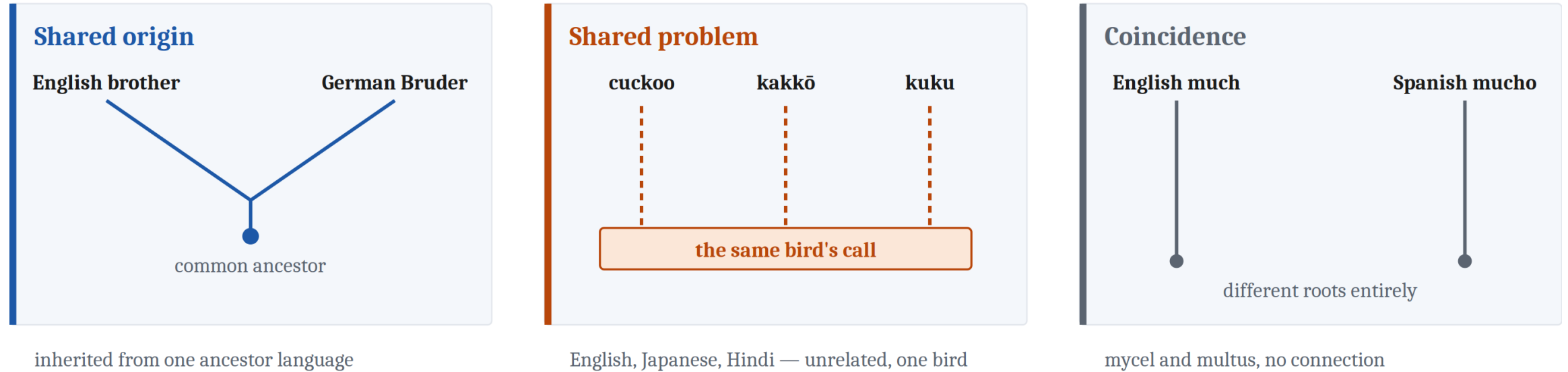


362 of the 400 carry the same verdict — 90.5% agreement, between two systems that have never met.
An observed 95% is 4.5 points more than that. The surplus is the finding.

Two independent systems, each right 95% of the time, with random errors, will still agree about 90% of the time. But 95% agreement is about five points above what independence alone produces. It is this surplus that calls for an explanation, not the agreement. Noticing that two things are similar is the easy part. The work is figuring out why.

Question One: Where Did the Similarity Come From?

Where did the similarity come from?



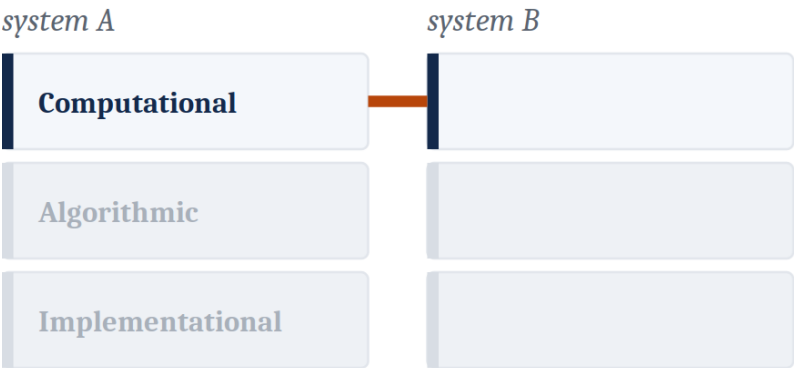
Only the first lets us reason about an ancestor. The second says there was none — which is a finding, not a consolation prize.

Shared origin lets us reason about the ancestor. Shared problem licenses **no** claim about an ancestor at all — arriving twice independently *is* the claim that the ancestor lacked it. Coincidence teaches nothing.

Question Two: How Deep Does It Go?

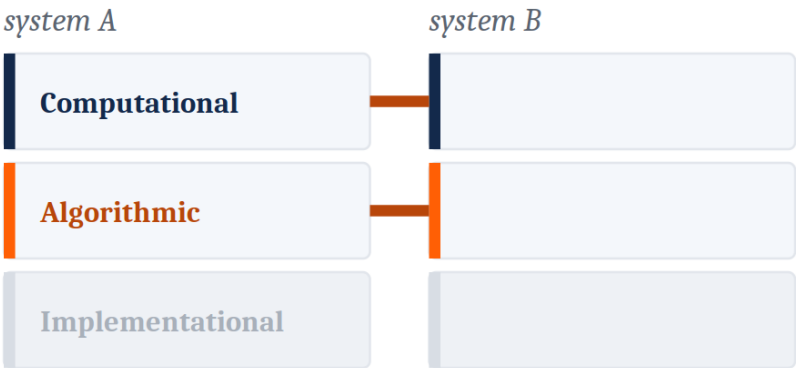
How deep does the similarity go?

The answer names the level at which the match holds.



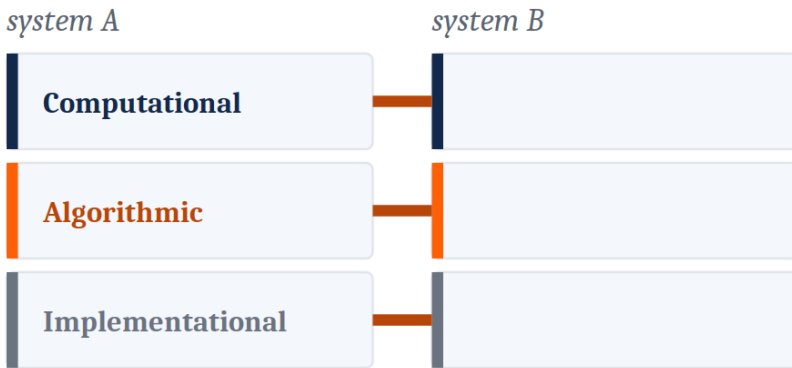
Behavior only

same output, different machinery



Down to the algorithm

same procedure, different stuff



All the way down

same procedure in the same stuff

And the reverse case is just as real: two systems can be built from nearly the same materials and be doing entirely different jobs with them.

A merely behavioral match teaches less than we might think.

And the reverse case is just as real: two systems can be built from nearly the same materials and be doing **entirely different jobs** with them.

Question Three: How Precisely Can We State It?

How precisely can we state it?

One behavioral claim, sharpened. Each step is the same finding, looked at harder.

both tell the two
categories apart

both do it at about
the same rate

both miss the same
kinds of item

same errors, same items,
same proportions

loose

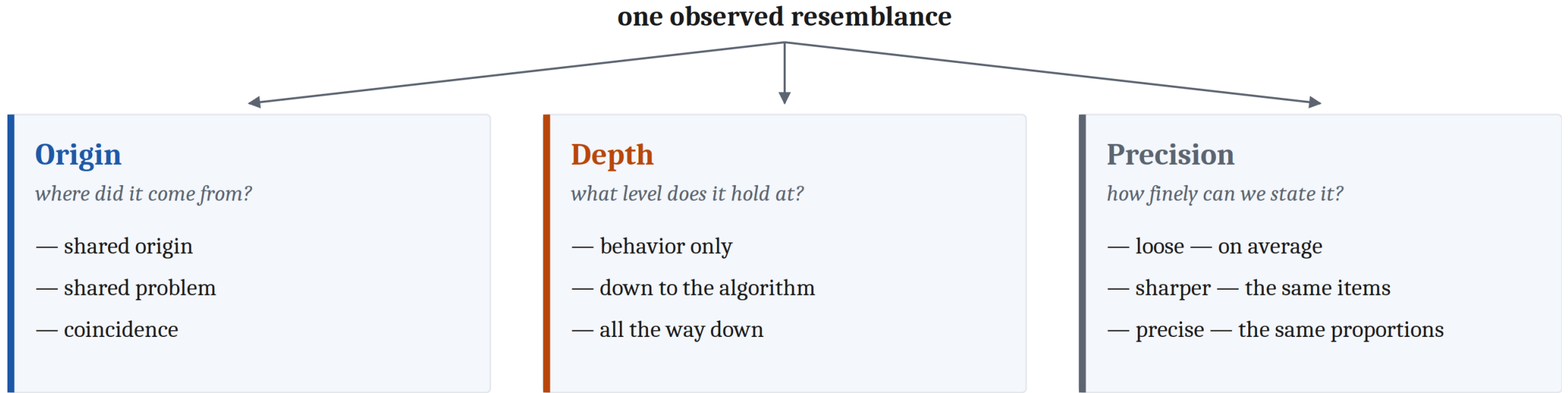
precise

Precision is not yes or no. It is how finely we looked before we stopped looking — and looking harder tends to surface differences.

Depth asks **at which level**. Precision asks **how finely specified** the match is at whatever level we are looking — so a claim can be **shallow and precise**, or **deep and vague**.

To say two systems behave alike is never to say they would behave alike in **every** circumstance. It is to say they matched on what we happened to measure.

Three Questions, Not Three Steps



Three questions, not three steps. They cross, they do not nest — and a shared mechanism is itself a resemblance.

They **cross**. They do not nest, and they are not a sequence.

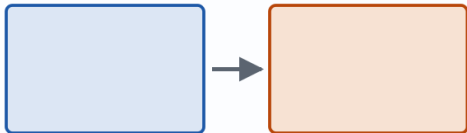
And a shared mechanism is **itself a resemblance** — which sends us straight back to question one.

Three Families of Comparison

Three families of comparison

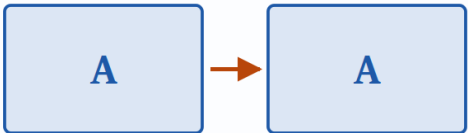
Different systems

species · ages · typical and atypical · machines



One system, two conditions

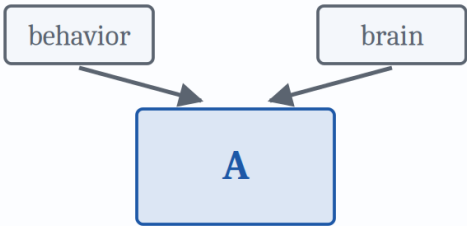
the difference chosen on purpose — the experiment



we make the difference

Two methods, one system

not why they are alike, but whether they agree



The honest asymmetry

what the field mostly does



what this course mostly teaches



■ different systems ■ conditions ■ methods

The same three questions run through all three families. Only the vocabulary changes.

Four Ways This Goes Wrong

Anthropomorphism — projecting human qualities onto a system that has not earned the description. The dog that looks guilty. The chatbot that sounds like it means it.

Anthropocentrism — treating the human case as the reference point, the default, the "normal" one.

Unconstrained adaptationism — assuming every trait must be a finely tuned solution to some problem.



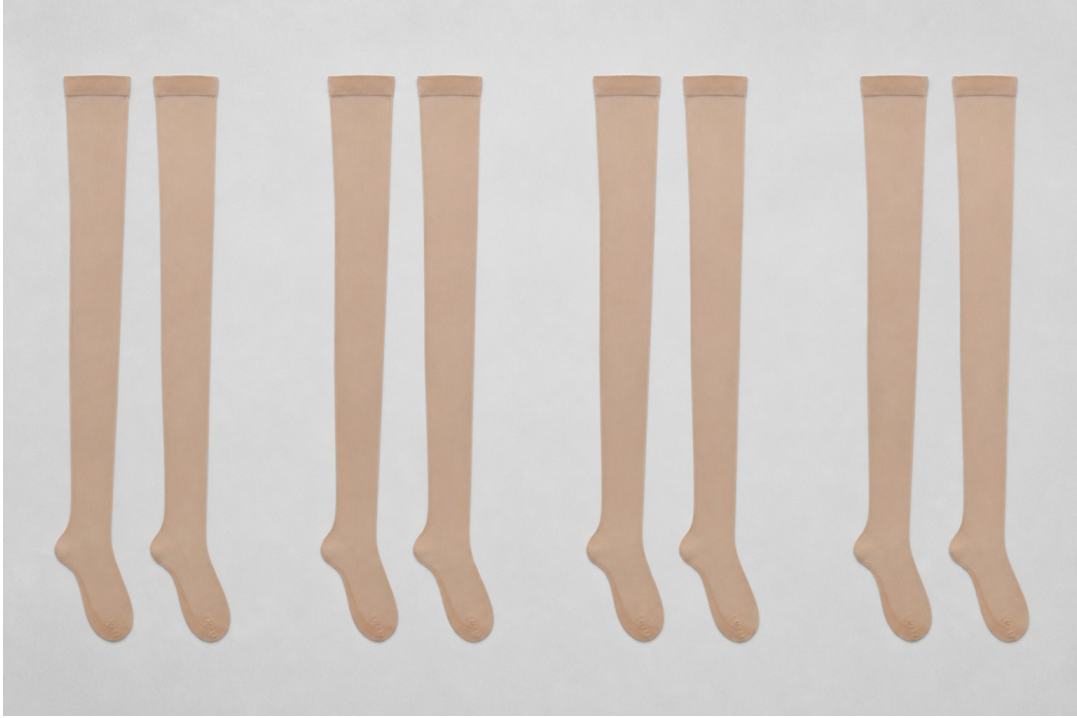
Four Ways This Goes Wrong



An error about depth and precision — and the one underneath the other three.

Surface matching is the theme under the other three. It is an error about **depth** and **precision**: two systems that behave alike get treated as alike underneath, when the only things checked were the handful of dimensions somebody happened to measure.

And We Do It to Ourselves



Nisbett and Wilson. Shoppers were shown several pairs of stockings laid out in a row.

The pairs were **identical**.

Shoppers chose the **rightmost** pair far more often than any other.

Asked why, they explained their choice by the **knit and the sheerness of the fabric** — and denied that position had anything to do with it.

We often have no direct access to why we did something. We generate a plausible reason instead.

None of This Is Neutral

Arguments about whether animals think, whether a fetus has experiences, whether a machine understands anything, are very often powered by something other than the evidence.

Claims about **who has which capacity** have historically tracked claims about **who is owed what**.

The pull runs both ways. Wanting to raise an animal's standing is a reason to read its behavior generously. Not wanting to change how animals are used is a reason to read the same behavior thinly.

Neither motive makes anyone wrong — but motives shape which studies get run, which results feel convincing, and which get explained away.

Also in the chapter, and still fair game: comparison across conditions (the experiment, confounds, control conditions) · across ages · across typical and atypical minds.

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2	Decomposition runs down one system; comparison runs across to another	fig-two-methods.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
3	Marr's three questions, asked of human and frog vision	fig-levels-vision.png	CC BY 4.0	Jon Willits — drawn for this lecture from the Module 2 reading's table · own work
4	Two wolves	Black_and_White_Wolves.jpg	CC BY 2.0	Stéfan, via Wikimedia Commons · https://commons.wikimedia.org/wiki/File:Black_and_White_Wolves.jpg
5	Two independent 95% verdicts over the same 400 photographs	fig-agreement-overlap.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
6	The three answers to the origin question, drawn as topology	fig-origin-topology.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
7	A match that stops at the top, in the middle, and all the way down	fig-depth-levels.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
8	One behavioral claim, sharpened four times	fig-precision-ladder.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
9	The three questions to ask of a resemblance	fig-three-questions.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work

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10	The three families, and the asymmetry between what the field does and what this course teaches	fig-three-families.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
11	A dog with a guilty expression	guilty_dog.png	Status unsettled; see this folder's IMAGE_CREDITS.md	Jon Willits — generated image · own work, generated with ChatGPT Codex
11	The three named errors, and the unnamed one underneath them	fig-surface-matching.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
12	Four identical pairs of stockings in a row	stocking_pairs.png	Status unsettled; see this folder's IMAGE_CREDITS.md	Jon Willits — generated image · own work, generated with ChatGPT Codex